

SDLC & Agile Scrum - Quick Revision Cheat Sheet

QUICK REVISION & EXAM CHEAT SHEET

Subject: Software Engineering | Chapter: Chapter 1: Software Development Life Cycles & Agile Methodologies | Academic Year: 2026-2027

SDLC Models Quick Comparison

- * Waterfall: Strict sequential flow. Best for fixed, well-understood requirements. High risk of late surprises.
- * V-Model: Verification and Validation phases pair directly with development phases.
- * Spiral: Best for high-risk, mission-critical systems. Heavy emphasis on risk analysis.
- * Agile: Iterative, incremental, welcomes changing requirements through short feedback loops.

Scrum Artifacts & Ceremonies

- * Artifacts: Product Backlog, Sprint Backlog, Product Increment (Definition of Done).
- * Ceremonies: Sprint Planning, Daily Standup (15m), Sprint Review (Demo), Sprint Retrospective.
- * Story Points measure relative complexity; Velocity measures points delivered per sprint.

Functional versus Non-Functional Requirements

- * Functional: System behavior, business logic (e.g., student can submit quiz, user can login).
- * Non-Functional: System qualities (e.g., API response < 200ms, 99.9% uptime, GDPR compliance).